

Roll No.

TPE-504

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
RESERVOIR ENGINEERING-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is non-Darcy flow ? Explain with the help of Darcy law for radial flow. (CO1)

OR

- (b) Illustrate dimensionless variables of the constant terminal rate solution of the radial diffusivity equation for transient state condition in terms of dimensionless pressure function. (CO1)

2. (a) Illustrate the constant terminal rate solution for the flow of a real gas. (CO1, CO2)

OR

- (b) A well is tested by producing it at a constant rate of 1500 stb/d for a period of 100 hours. It is suspected, from seismic and geological evidence, that the well is draining an isolated reservoir block which has

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approximately a 2 : 1 rectangular geometrical shape and the extended drawdown test is intended to confirm this. This reservoir data and flowing bottom hole pressures recorded during the test are detailed below and in table. (CO1, CO2)

$$h = 25 \text{ ft}, c = 15 \times 10^{-6} / \text{psi}, \text{Porosity} = .20, B_o = 1.25 \text{ rb/stb},$$

$$r_w = .43 \text{ ft}, \text{Viscosity} = 1 \text{ CP}$$

Flowing Time (hours)	P_{wf} (psia)	Flowing time (hours)	P_{wf} (psia)
0	3500 (p_i)	20	2762
1	2917	30	2703
2	2900	40	2650
3	2888	50	2597
4	2879	60	2545
5	2869	70	2495
7.5	2843	80	2443
10	2830	90	2392
15	2794	100	2341

(i) Calculate the effective permeability of well.

(ii) Calculate skin factor of the well.

3. (a) Express t_D , in field units with the real time in hours and days and using dimensional analysis, show that both t_D , and p_D are dimensionless.

(CO1, CO2)

(3)

OR

- (b) What is general theory of well testing ? Explain. (CO1, CO2)
4. (a) Explain the concept of constant terminal rate solution for transient state. (CO1)

OR

- (b) Determine the pressure response at the wellbore due to the production of a well which is situated at the centre of a square drainage area with sides measuring : (CO1)
- (i) $L = 100$ ft
- (ii) $L = 500$ ft
- The relevant reservoir and fluid properties are as follows $k = 50$ mD, Porosity = 0.3, Viscosity = 1 cp, $c = 15 \times 10^{-6}$ / psi, $r_w = 0.3$ ft, CA = 30.9.
5. (a) Illustrate basic equation of radial flow of a single-phase fluid and basic differential equation for radial fluid flow. (CO1, CO2)

OR

- (b) Using diffusivity equation prove dimensionless pressure. (CO1, CO2)